

Notes: Edelen, Ince, and Kadlec (JFE, 2016)

Institutional Investors and Stock Return Anomalies

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1 Big picture

This paper asks a direct question about institutional investors: when stocks take on characteristics that predict anomaly returns, do institutions trade in the direction suggested by the anomalies, or do they trade against them?

The standard intuition in asset pricing is that institutions should be relatively sophisticated arbitrageurs. If stock-return anomalies capture mispricing or benchmark-adjusted return opportunities, institutional investors should tilt toward long-leg stocks and away from short-leg stocks. Edelen, Ince, and Kadlec find the opposite in many cases. Institutions tend to buy stocks classified as overvalued by anomaly signals, especially the short legs of the anomaly portfolios, and those same stocks later earn particularly negative abnormal returns.

The paper is important because it changes the role of institutions in the anomaly debate. Instead of treating institutions mainly as arbitrageurs who are limited by frictions, the evidence suggests that institutional demand itself can be associated with persistent mispricing.

2 Research question and hypothesis

2.1 Core research question

Do institutional investors exploit well-known stock return anomalies?

The authors examine this through two linked tests:

1. **Trading direction.** As anomaly characteristics form, do institutions buy long-leg stocks and sell short-leg stocks?
2. **Return informativeness.** Conditional on anomaly characteristics, do institutional trades predict better future returns, or do they predict worse returns?

2.2 Sophisticated institutions hypothesis

The paper frames the analysis around the **sophisticated institutions hypothesis** (SIH):

- If institutions are sophisticated arbitrageurs, their trades should align with anomaly prescriptions.
- If institutions trade against anomaly prescriptions, their stock selection should at least offset the anomaly signal. For example, if they buy a short-leg stock, that stock should not go on to earn especially bad returns.

The main findings reject both parts of this hypothesis in the aggregate data.

3 Incremental contribution of the paper

1. **Moves from institutional holdings to institutional changes.** Lewellen (2011) shows that aggregate institutional holdings do not strongly tilt toward anomalies. This paper focuses on

changes in institutional ownership during the period when anomaly characteristics are formed, which is closer to observing active portfolio decisions.

2. **Uses the institutional-investor base, not only percent ownership.** The main measure is the change in the number of institutions holding a stock. This gives an equal-weighted view of institutional entry and exit and is less dominated by a few large managers than percent ownership.
3. **Links institutional demand to both anomaly-predicted and residual returns.** The paper shows institutions load negatively on anomaly characteristics and that their residual trading also predicts lower future returns.
4. **Reconciles conflicting evidence on institutional skill.** Prior studies often find that institutional buying predicts positive returns over short horizons. This paper shows that the relation is positive over short horizons but turns negative over the one-year horizon relevant for anomaly tests.
5. **Challenges the standard limits-of-arbitrage story.** The strongest anomaly losses occur in overvalued short-leg stocks that institutions buy rather than avoid. This weakens explanations based only on short-sale constraints, transaction costs, or institutions' unwillingness to bear idiosyncratic risk.
6. **Provides evidence against flow as the main mechanism.** Results survive controls for mutual fund ownership, flow-induced mutual fund trades, and samples without flow pressure.
7. **Suggests an agency or preference-based mechanism.** Institutions may prefer stocks that look like "good companies" even when those characteristics forecast poor long-run returns.

4 Data and construction of key variables

4.1 Sample and data sources

The empirical sample covers U.S. common stocks traded on NYSE, Amex, and Nasdaq. The authors use CRSP returns, Compustat accounting variables, Thomson-Reuters 13F institutional holdings, mutual fund holdings, and SDC issuance data. The main anomaly-return sample runs from July 1982 through June 2012, with institutional holdings data beginning in December 1980 and ending in June 2011.

The authors exclude utilities, financial firms, and stocks priced below \$5 at portfolio formation. Returns are adjusted for delisting bias. Anomaly portfolios are value-weighted, but the analysis of institutional ownership changes is generally based on equal-weighted observations.

4.2 Seven anomalies

The paper studies seven broad anomaly categories:

Anomaly	Label	Economic idea
Net operating assets	NOA	Accumulated operating accruals and balance-sheet bloat predict low future returns.
Gross profitability	GP	More profitable firms earn higher future returns.
Investment to assets	IVA	Firms with high investment growth tend to underperform.
O-Score	OSC	Distress characteristics predict abnormal returns.
Book-to-market	B/M	Value firms outperform growth firms.
Momentum	MOM	Recent winners outperform recent losers over intermediate horizons.
Undervalued minus overvalued	UMO	Repurchases versus issuances identify undervaluation versus overvaluation.

4.3 Institutional demand measure

The main institutional demand measure is the change in the number of institutions holding a stock during the anomaly-formation window:

$$\Delta \#Inst_i = \frac{\#Inst_{i,q} - \#Inst_{i,q-5}}{\text{average beginning-period } \#Inst \text{ for stocks in the same market-cap decile}}.$$

This scaling avoids problems when the initial number of institutional holders is zero or very small. The authors also study percent institutional ownership, percent mutual fund ownership, non-mutual-fund ownership, and flow-induced mutual fund trades.

4.4 Timing

For accounting-based anomalies, portfolios are formed at the end of June in year t , using accounting information from fiscal year $t - 1$. The anomaly return window is July of year t through June of year $t + 1$. Institutional trading is measured over the six quarters ending at portfolio formation, from $q - 5$ to q . Momentum is formed quarterly because its signal is more time-sensitive.

5 Empirical strategy

5.1 Portfolio-level tests

The authors first sort stocks into anomaly long, neutral, and short legs. They then measure institutional buying during the formation window. The key test is whether institutional demand is higher in the long leg than the short leg.

They then independently sort stocks on anomaly characteristics and institutional demand. This creates four important portfolios:

- long-leg stocks institutions buy;

- long-leg stocks institutions sell;
- short-leg stocks institutions buy;
- short-leg stocks institutions sell.

If institutions are sophisticated arbitrageurs, the short-leg-buy portfolio should not have especially poor returns. In fact, it has the worst abnormal returns.

5.2 Return measurement

Returns are reported as excess returns and Fama-French three-factor alphas. For the book-to-market anomaly, the HML factor is excluded from the factor model to avoid mechanically controlling away the value anomaly.

5.3 Fama-MacBeth regressions

The paper also estimates monthly cross-sectional regressions of future stock returns on institutional ownership changes, anomaly characteristics, size, and illiquidity. A representative specification is:

$$R_{i,m} = a_m + b_m \Delta \#Inst_{i,q-5:q-1} + c_m \Delta \#Inst_{i,prior\ quarter} + \Gamma_m X_i + \varepsilon_{i,m},$$

where X_i includes the seven anomaly characteristics, log market capitalization, and Amihud illiquidity.

This decomposition separates:

- the long-horizon institutional demand effect;
- the short-horizon institutional demand effect;
- the anomaly-characteristic component of future returns;
- residual returns not explained by anomaly characteristics.

6 Main findings

6.1 Finding 1: institutions do not trade like anomaly arbitrageurs

Institutions generally do not buy long-leg stocks and sell short-leg stocks. They often do the reverse. Across the seven anomalies, short-leg stocks experience larger increases in institutional ownership than long-leg stocks.

This is especially strong for net operating assets, investment-to-assets, book-to-market, and undervalued-minus-overvalued. Momentum is the main exception: institutions buy winners and sell losers, which aligns with the momentum anomaly.

6.2 Finding 2: overvalued stocks bought by institutions underperform most

The strongest abnormal returns come from short-leg stocks that institutions buy. Across anomalies, short-leg stocks bought by institutions earn an average alpha of roughly -0.74% per month, compared with about -0.27% per month for short-leg stocks institutions sell.

This finding is central. It shows that institutions are not simply identifying short-leg stocks that will defy the anomaly. Instead, when they buy short-leg stocks, those stocks perform especially badly.

6.3 Finding 3: institutional trades persist

Institutions do not immediately reverse their pre-anomaly trades. The paper shows that changes in institutional investor base persist through the anomaly return window. This matters because institutional investors are exposed to the subsequent long-run underperformance.

6.4 Finding 4: the institutional-demand return relation changes with horizon

Short-horizon institutional demand predicts positive near-term returns. This is consistent with prior literature. But long-horizon demand predicts negative returns over the one-year horizon used in anomaly research.

The authors argue that this horizon result reconciles their evidence with earlier findings that institutional trading is positively related to future returns.

6.5 Finding 5: flow is not the main explanation

The paper tests whether mutual fund flows drive the results. The evidence does not support this. The negative relation between institutional demand and future returns survives:

- using changes in the number of institutions rather than percent ownership;
- decomposing mutual fund and non-mutual-fund ownership;
- controlling for flow-induced mutual fund trading;
- excluding stocks under strong flow pressure.

6.6 Finding 6: conventional limits-of-arbitrage explanations are incomplete

If anomalies persist because institutions face short-sale constraints, one would expect institutions to avoid buying overvalued short-leg stocks. The evidence shows the opposite. Institutions increase their holdings of many short-leg anomaly stocks.

This suggests that the key arbitrage risk may not be just short-sale constraints or transaction costs. Instead, arbitrageurs may be reluctant to short against widespread and persistent institutional buying.

7 Interpretation and economic mechanism

The paper does not claim that every institutional investor is unsophisticated. It studies aggregate institutional demand, and some institution types may still exploit anomalies. The main result is that the aggregate institutional sector does not appear to discipline these anomalies.

The authors discuss several mechanisms:

- **Investor flow.** Largely ruled out by the flow tests.
- **Short-sale constraints and transaction costs.** These may matter, but they cannot fully explain why institutions actively buy overvalued stocks.
- **Biased expectations.** Institutional buying is negatively related to future earnings-announcement returns, suggesting that some trades may reflect overly optimistic cash-flow expectations.
- **Agency and “good company” preferences.** Managers may prefer stocks that are easier to justify to clients, such as growing firms, glamorous firms, or firms with superficially attractive characteristics, even when those characteristics forecast poor returns.

8 What to remember

- The paper documents a negative long-horizon relation between institutional demand and future anomaly returns.
- Institutions often buy the short leg of anomalies, especially stocks classified as overvalued.
- These short-leg stocks bought by institutions earn particularly negative abnormal returns.
- Prior positive evidence on institutional demand and returns is mostly a short-horizon phenomenon.
- The results challenge the view that institutions simply arbitrage away anomaly mispricing.
- The paper suggests that institutional demand itself can be part of the mispricing process.

9 Original tables and figures with interpretation

9.1 Table 1: Anomalies considered; Table 2: Anomaly returns

Anomaly	Label	Description	Citation
Net operating assets	NOA	The sum of short-term debt (DLC), long-term debt (DLTT), minority interest (MBI), preferred stock (PSTK), and common equity (CEQ) minus cash and short-term investment (CHE), deflated by lagged total assets (AT).	Hirshleifer, Hsu, Teoh, and Zhang (2004)
Gross profitability	GP	Total revenues (REVTT) minus cost of goods sold (COGS), divided by total assets (AT).	Novy-Marx (2013)
Investment to assets	IVA	The change in gross property, plant, and equipment (PPEGT) plus the change in inventories (INVT), deflated by the lagged total assets (AT).	Lyandres, Sun, and Zhang (2008)
O-Score	OSC	The probability of bankruptcy calculated using accounting variables such as total liabilities divided by assets, working capital divided by assets, current liabilities divided by current assets, net income, and inflation-adjusted total assets applied to coefficients estimated using a logit regression of bankruptcies.	Ohlson (1980); Dichev (1998)
Book to market	B/M	Book value of common equity (SQE or AT-LT) plus net deferred tax assets (TXDB), investment tax credit (ITCR), and post-retirement benefit liabilities (PRBA), divided by market capitalization at end of calendar year t .	Fama and French (1992)
Momentum	MOM	Cumulative stock return between months $j-1$ and $j-12$, where $j+1$ to $j+3$ are the months of performance evaluation.	Jegadeesh and Titman (1993)
Undervalued minus overvalued	UMO	The portfolio "U" (undervalued) contains firms with equity or debt repurchases and without any equity or debt issuances during the two most recent fiscal years. The portfolio "O" (overvalued) contains firms with equity or debt issuances and without any equity or debt repurchases during the two most recent fiscal years.	Hirshleifer and Jiang (2010)

Table 1: Anomalies considered

Table 2
Anomaly returns.
The table presents average monthly returns in percent for the seven anomaly portfolios in Table 1 and their equal-weighted overall average (last column), between July of 1982 and June of 2012. Anomaly portfolios are formed by ranking on the indicated variable as of June year t and taking a long (short) position in the highest (lowest) performing 30% tails. For all anomalies except momentum (see Table 1 for acronyms), the position is held July year t through June of year $t+1$. Momentum portfolios are rebalanced quarterly using prior 12-month stock returns. Excess returns (Panel A) net out the one-month US Treasury bill rate; three-factor alphas (Panel B) refer to the intercept from a time-series regression of monthly value-weighted excess returns on the MKT, SMB, and HML factors, excluding HML for the B/M anomaly. Heteroskedasticity-adjusted t -statistics are in parentheses.

	NOA	GP	IVA	OSC	B/M	MOM	UMO	AVG
<i>Panel A: Monthly excess returns (%)</i>								
Long leg	0.86	0.96						
Short leg	0.32	0.35	0.37	0.56	0.35	0.43	0.40	0.40
Long-short	0.53	0.61	0.52	0.20	0.59	0.47	0.57	0.50
	(4.7)	(5.2)	(5.0)	(1.8)	(3.3)	(1.9)	(5.1)	(6.3)
<i>Panel B: Monthly three-factor alphas (%)</i>								
Long leg	0.17	0.28	0.12	0.10	0.26	0.20	0.25	0.19
	(1.9)	(3.5)	(1.8)	(1.5)	(2.2)	(2.1)	(3.7)	(2.0)
Short leg	-0.50	-0.49	-0.41	-0.30	-0.46	-0.50	-0.38	-0.41
	(-5.2)	(-5.0)	(-4.4)	(-3.0)	(-4.3)	(-2.6)	(-3.9)	(-3.3)
Long-short	0.67	0.76	0.53	0.40	0.72	0.70	0.63	0.60
	(6.5)	(6.8)	(6.0)	(4.5)	(4.0)	(3.2)	(6.8)	(8.6)

Table 2: Anomaly returns

How to read Table 1. The paper deliberately uses a compact set of seven anomalies rather than every published anomaly. The seven signals span valuation, profitability, investment, earnings quality, distress, momentum, and financing. This design makes the test broad but avoids overloading the analysis with highly correlated anomalies.

How to read Table 2. The anomalies are present in this sample. The average long-short three-factor alpha across the seven anomalies is about 0.60% per month. The table also confirms a common feature of anomaly returns: abnormal performance is concentrated in the short leg, where the average short-leg alpha is about -0.41% per month.

9.2 Table 3: Summary statistics for institutional ownership; Figure 1: Time line for the construction of anomaly portfolios

Table 3

Summary statistics for institutional ownership.

Panel A reports the means, standard deviations, and percentiles for the level of and change in the number of 13F institutional investors ($\#Inst$) and the percentage of shares held by 13F institutional investors ($\%Inst$) during the six calendar quarters prior to annual portfolio formation at the end of June ($q-5$ to q), where q refers to end of June of each year beginning in 1982. All variables are winsorized at the 1% level in both tails. Change variables are converted to annualized equivalent values. Panel B reports the means for quintiles sorted annually by $\Delta\#Inst$ ($q-5$ to q). Data are from December 1980 through June 2011.

Panel A: Summary statistics					
	Mean	Standard deviation	P25	Median	P75
$\#Inst$ (q)	84.6	118.1	13.0	40.0	108.0
$\%Inst$ (q)	42.8%	28.8%	17.8%	40.2%	65.4%
$\Delta\#Inst$ ($q-5$ to q)	13.3%	25.9%	0.0%	10.6%	27.0%
$\Delta\%Inst$ ($q-5$ to q)	4.7%	10.1%	-0.4%	2.5%	7.8%
Panel B: Means for portfolios sorted by $\Delta\#Inst$ ($q-5$ to q)					
	Large decrease	2	3	4	Large increase
$\#Inst$ (q)	66.4	104.8	114.4	92.2	61.9
$\%Inst$ (q)	41.8%	43.6%	45.9%	45.8%	45.3%
$\Delta\#Inst$ ($q-5$ to q)	-19.2%	1.8%	12.3%	24.2%	50.3%
$\Delta\%Inst$ ($q-5$ to q)	-1.3%	1.2%	2.6%	5.2%	15.1%

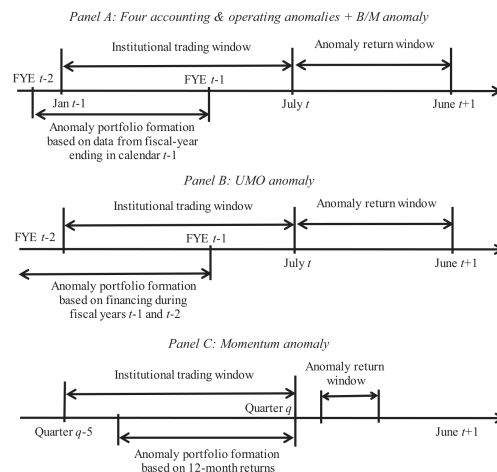


Fig. 1. Time line for the construction of anomaly portfolios. Fig. 1 depicts the portfolio construction time line for the seven stock return anomalies of Table 1. For all anomalies except momentum, portfolios are constructed annually at the end of June of each year t based on stock characteristics as of fiscal year-end in year $t-1$ and changes in institutional ownership during the previous six calendar quarters ($q-5$ to q), and held for the next twelve months. For momentum, portfolios are constructed quarterly based on stock returns during the previous 12 months and changes in institutional ownership during the previous six calendar quarters ($q-5$ to q), and held for the next three months. FYE $t-1$ indicates the fiscal year-end in calendar year $t-1$.

Table 3: Summary statistics for institutional ownership

Figure 1: Time line for the construction of anomaly portfolios

How to read Table 3. Institutional ownership is widespread: the average stock has many institutional holders and a large percentage of shares held by institutions. The table also shows substantial variation in changes in institutional ownership, which is what allows the paper to test whether institutional demand is related to anomaly signals.

How to read Figure 1. The key institutional trading window is the six-quarter period before anomaly returns are measured. This timing is central: the authors want to capture how institutions trade while anomaly characteristics are becoming visible and before anomaly returns are realized.

9.3 Table 4: Changes in institutional investor base for anomaly stocks; Table 5: Abnormal returns of anomaly portfolios conditional on institutional demand

Table 4
Changes in institutional investor base for anomaly stocks.
The table presents annualized changes in institutional ownership during the anomaly portfolio formation period, 1982–2012. [See Fig. 1 for timeline and Table 1 for anomaly acronyms.] $\Delta\#Inst$ is the change in the number of institutional investors scaled by the average number of institutions holding stocks in the same market capitalization decile at the beginning of the period. $\Delta\%Inst$ is the change in fraction of shares held. +, ++, and + indicate p -values of 1%, 5%, and 10% or less, respectively. The p -values are estimated nonparametrically and adjusted for autocorrelation using moving block bootstrapping.

		$\Delta\#Inst$	$\Delta\%Inst$			$\Delta\#Inst$	$\Delta\%Inst$
NOA	Long	12.2%	4.3%	GP	Long	15.0%	4.5%
	Neut	10.7%	3.3%		Neut	12.5%	3.9%
	Short	17.2%	5.3%		Short	12.0%	4.4%
	L-N	1.5%	1.0%		L-N	2.5%	0.6%
	S-N	6.5%***	2.0%***		S-N	-0.5%	0.5%
IVA	L-S	-5.0%***	-1.0%	OSC	L-S	3.0%*	-0.1%
	Long	9.3%	3.2%		Long	13.6%	4.3%
	Neut	11.6%	3.7%		Neut	12.1%	3.8%
	Short	17.3%	5.4%		Short	12.5%	4.2%
	L-N	-2.3%	-0.5%		L-N	1.5%	0.5%
BM	S-N	5.7%***	1.7%***	MOM	S-N	0.3%	0.4%
	L-S	-8.0%***	-2.2%***		L-S	1.2%	0.1%
	Long	5.6%	2.3%		Long	19.7%	5.6%
	Neut	12.6%	3.9%		Neut	9.5%	3.1%
	Short	20.9%	6.4%		Short	4.0%	2.0%
UMO	L-N	-7.0%***	-1.6%*	AVG	L-N	10.2%***	2.5%***
	S-N	8.3%***	2.5%***		S-N	-5.4%***	-1.1%***
	L-S	-15.4%***	-4.1%***		L-S	15.7%***	3.6%***
	Long	9.4%	2.4%		Long	12.1%	3.8%
	Neut	11.0%	3.6%		Neut	11.4%	3.6%
	Short	21.6%	7.5%		Short	15.1%	5.0%
	L-N	-1.6%	-1.2%*		L-N	0.7%	0.2%
	S-N	10.6%***	3.9%***		S-N	3.7%***	1.4%***
	L-S	-12.2%***	-5.1%***		L-S	-3.0%***	-1.2%***

Table 4: Changes in institutional investor base for anomaly stocks

Table 5
Abnormal returns of anomaly portfolios conditional on institutional demand.
The table presents monthly three-factor alphas during the anomaly return window sorted independently by anomaly characteristics and change in institutional ownership during the anomaly portfolio formation period. [See Fig. 1 for timeline and Table 1 for anomaly acronyms.] Buy (Sell) refers to the top (bottom) quintile of change in institutional ownership. Panel A presents seven anomaly characteristics separately, using $\Delta\#Inst$ to measure change in institutional ownership (change in number of institutional investors scaled by the average number of institutions holding stocks in the same market capitalization decile at the beginning of the period). Panel B presents an equal-weighted combination of the seven anomalies using $\Delta\#Inst$, $\Delta\%Inst$, $\Delta\%MF$, and $\Delta\%MF$ to measure change in institutional ownership, where “ $\Delta\%$ ” refers to the change in percentage of shares held and “MF” refers to mutual funds. Heteroskedasticity-adjusted t -statistics are in parentheses.

Panel A: Portfolio abnormal returns for seven anomalies using $\Delta\#Inst$														
	MOA		B-S		GP		B-S		I/M		OSC		B-S	
	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell
Long	0.23 (1.6)	0.20 (1.3)	0.03 (0.2)	0.12 (1.2)	0.35 (2.6)	-0.19 (-1.2)	-0.06 (-0.5)	0.14 (1.2)	-0.20 (-1.3)	-0.14 (-1.0)	0.23 (1.7)	-0.37 (-2.3)	0.23 (1.7)	-0.37 (-2.3)
Short	-0.75 (-4.6)	-0.15 (-1.0)	-0.60 (-3.1)	-0.82 (-4.6)	-0.44 (-3.2)	-0.38 (-2.0)	-0.72 (-4.7)	-0.28 (-1.8)	-0.44 (-2.5)	-0.65 (-4.1)	-0.26 (-1.6)	-0.39 (-2.0)	-0.26 (-1.6)	-0.39 (-2.0)
L-S	0.98 (6.0)	0.35 (2.2)	0.63 (2.8)	0.98 (5.3)	0.79 (4.8)	0.19 (0.9)	0.66 (4.5)	0.42 (2.8)	0.24 (1.3)	0.51 (2.5)	0.49 (2.8)	0.02 (0.1)	0.49 (2.8)	0.02 (0.1)
	BM		MOM		MOM		OMO		OMO		OMO		OMO	
	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell
Long	0.09 (0.5)	0.35 (2.1)	-0.26 (-1.5)	0.18 (1.4)	0.07 (0.4)	0.11 (0.6)	0.47 (2.8)	0.06 (0.4)	0.06 (0.4)	0.06 (0.4)	0.06 (0.4)	0.06 (0.4)	0.06 (0.4)	0.06 (0.4)
Short	-0.74 (-4.2)	-0.19 (-1.1)	-0.55 (-2.1)	-1.15 (-4.4)	-0.32 (-1.5)	-0.83 (-4.2)	-0.61 (-4.2)	-0.61 (-4.2)	-0.20 (-1.2)	-0.20 (-1.2)	-0.20 (-1.2)	-0.20 (-1.2)	-0.20 (-1.2)	-0.20 (-1.2)
L-S	0.83 (3.1)	0.54 (3.1)	0.29 (1.2)	1.43 (4.8)	0.39 (1.8)	0.94 (3.4)	0.98 (5.7)	0.35 (1.3)	0.73 (3.0)	0.73 (3.0)	0.73 (3.0)	0.73 (3.0)	0.73 (3.0)	0.73 (3.0)
Panel B: Equal-weighted average portfolio abnormal returns across seven anomalies														
	$\Delta\#Inst$		$\Delta\%Inst$		$\Delta\%MF$		$\Delta\%MF$		$\Delta\%MF$		$\Delta\%MF$		$\Delta\%MF$	
	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell	Buy	Sell
Long	0.10 (1.0)	0.16 (1.5)	-0.06 (-0.5)	0.02 (0.3)	0.23 (3.0)	-0.21 (-2.3)	0.04 (0.4)	0.21 (2.2)	-0.17 (-1.6)	0.07 (0.9)	0.19 (2.1)	-0.12 (-1.3)	0.19 (2.1)	-0.12 (-1.3)
Short	-0.74 (-5.0)	-0.27 (-1.9)	-0.47 (-3.3)	-0.62 (-4.7)	-0.35 (-2.6)	-0.27 (-2.5)	-0.58 (-4.2)	-0.34 (-2.6)	-0.24 (-1.9)	-0.40 (-3.5)	-0.42 (-3.5)	0.02 (0.2)	-0.42 (-3.5)	0.02 (0.2)
L-S	0.84 (8.0)	0.43 (5.4)	0.41 (4.0)	0.64 (7.3)	0.58 (6.0)	0.06 (0.8)	0.62 (6.6)	0.55 (6.9)	0.07 (0.7)	0.47 (6.2)	0.61 (7.3)	-0.14 (-1.6)	0.61 (7.3)	-0.14 (-1.6)

Table 5: Abnormal returns of anomaly portfolios conditional on institutional demand

How to read Table 4. Institutions do not behave like anomaly arbitrageurs. Across anomalies, short-leg stocks often experience larger increases in institutional ownership than long-leg stocks. The average short-leg change in the institutional investor base is higher than the average long-leg change.

How to read Table 5. The worst future performance occurs when institutions buy anomaly short-leg stocks. The result is not just that institutions fail to arbitrage anomalies; their demand is concentrated in the stocks that subsequently produce the strongest negative abnormal returns.

9.4 Table 6: Anomaly portfolio characteristics conditional on institutional demand; Table 7: Quarterly abnormal returns of anomaly portfolios

Table 6

Anomaly portfolio characteristics conditional on institutional demand.

Panel A presents averages across the seven anomalies for variables observed during the six quarters preceding the anomaly return window, 1982–2012. % of sample is the count of stocks in the portfolio divided by the total count of the four portfolios. $\Delta\#Inst$ is the scaled change in number of institutions; $\Delta\%Inst$ ($\Delta\%MF$) is the change in percentage of shares held by institutions (mutual funds); and $\Delta\#Inst$ next 4 qtrs is the change during the anomaly return window. In Panel B, $\#Inst$, $\%Inst$ and $MCap$ (market capitalization in millions 2012 dollars) are as of the beginning of year $t-1$. 'Idio. volatility' is the average annualized volatility of monthly three-factor residuals July $t-3$ through June t ; 'Amihud illiquidity' is the average monthly Amihud illiquidity ratio during year $t-1$ winsorized at 1%.

Panel A: Change in institutional ownership, averaged across the seven anomalies					
	% of sample	$\Delta\#Inst$	$\Delta\%Inst$	$\Delta\%MF$	$\Delta\#Inst$ next 4 qtrs
<i>Institutional buys:</i>					
Long leg	21.7%	81.9%	19.9%	3.0%	9.9%
Short leg	29.3%	85.8%	22.8%	3.6%	8.9%
[p-value for difference]	[0.000]	[0.481]	[0.060]	[0.171]	[0.721]
<i>Institutional sells:</i>					
Long leg	24.8%	-30.3%	-2.1%	0.0%	13.2%
Short leg	24.3%	-32.4%	-2.6%	-0.1%	10.6%
[p-value for difference]	[0.577]	[0.043]	[0.148]	[0.511]	[0.038]
Panel B: Stock characteristics, averaged across the seven anomalies					
	$\#Inst$	$\%Inst$	$MCap$ (2012 \$M)	Idio. volatility	Amihud illiquidity
<i>Institutional buys:</i>					
Long leg	64.4	46.1%	\$1,010	47.1%	0.15
Short leg	59.3	43.2%	\$963	50.4%	0.13
[p-value for difference]	[0.587]	[0.445]	[0.701]	[0.191]	[0.641]
<i>Institutional sells:</i>					
Long leg	63.6	41.6%	\$1,192	40.8%	0.25
Short leg	71.7	42.3%	\$1,428	42.9%	0.16
[p-value for difference]	[0.074]	[0.695]	[0.060]	[0.006]	[0.000]

Table 6: Anomaly portfolio characteristics conditional on institutional demand

Table 7

Quarterly abnormal returns of anomaly portfolios.

The table presents quarterly three-factor alphas during each of the four quarters of the anomaly return window, averaged across seven anomalies. Portfolios are constructed by independently sorting on anomaly characteristics and $\Delta\#Inst$, defined as the change in number of institutional investors scaled by the average number of institutions holding stocks in the same market capitalization decile at the beginning of the period. Buy-minus-Sell portfolios are long (short) stocks in the top (bottom) quintile of $\Delta\#Inst$; in each case restricted to stocks that are also in the long, or short, anomaly leg as indicated in the row heading. $\Delta\#Inst$ is measured during the six-quarter institutional trading window ($q-5$ to q) in Panel A; the five preceding quarters ($q-5$ to $q-1$) in Panel B; and the quarter prior to the anomaly return window (q) in Panel B. Heteroskedasticity-adjusted t -statistics are in parentheses.

	$q+1$	$q+2$	$q+3$	$q+4$	Cumulative
<i>Panel A: Buy-minus-sell defined using $\Delta\#Inst$ ($q-5$ to q)</i>					
Long leg	0.27	0.11	-0.34	-0.04	-0.06
	(1.0)	(0.5)	(-1.8)	(-0.2)	(-0.5)
Short leg	-0.14	-0.43	-1.01	-0.39	-0.47
	(-0.5)	(-1.2)	(-3.8)	(-1.7)	(-3.3)
<i>Panel B: Buy-minus-sell defined using $\Delta\#Inst$ (q)</i>					
Long leg	0.65	0.59	-0.13	0.12	0.25
	(3.2)	(2.5)	(-0.4)	(0.6)	(1.9)
Short leg	0.64	0.93	-0.07	0.29	0.40
	(2.5)	(2.7)	(-0.2)	(1.2)	(2.5)
<i>Panel C: Buy-minus-sell defined using $\Delta\#Inst$ ($q-5$ to $q-1$)</i>					
Long leg	0.23	-0.26	-0.44	-0.15	-0.19
	(1.0)	(-1.1)	(-1.7)	(-0.7)	(-1.7)
Short leg	-0.22	-0.69	-1.27	-0.58	-0.69
	(-0.8)	(-1.9)	(-3.9)	(-2.6)	(-4.6)

Table 7: Quarterly abnormal returns of anomaly portfolios

How to read Table 6. The table checks whether the conditional portfolios differ mechanically in obvious characteristics. Short-leg stocks bought by institutions do not look dramatically different from long-leg stocks bought by institutions on the main dimensions reported. This weakens simple explanations based only on size, illiquidity, or beginning institutional ownership.

How to read Table 7. The institutional demand-return relation is horizon dependent. Short-horizon institutional demand has a positive relation with near-term returns. But when the institutional trading window excludes the most recent quarter and focuses on the longer horizon, the buy-minus-sell spread becomes negative, especially for short-leg stocks.

9.5 Figure 2: Cumulative changes in number of institutional investors across seven anomalies; Table 8: Fama-MacBeth regressions of stock returns on institutional ownership changes

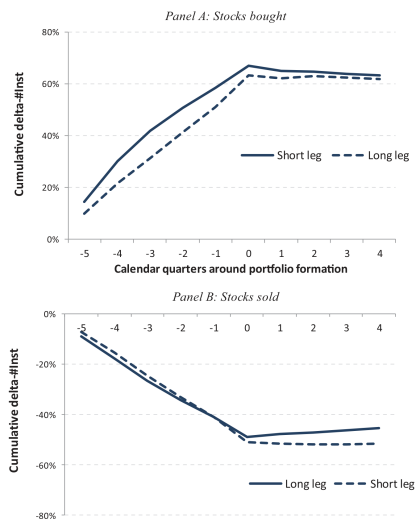


Fig. 2. Cumulative changes in number of institutional investors across seven anomalies. This chart depicts the average detrended $\Delta\#Inst$ across seven anomalies, cumulated over the portfolio formation window (quarters -5 to 0) through the anomaly return window (quarters 1-4). $\Delta\#Inst$ is the change in the number of institutional investors scaled by the average number of institutions holding stocks in the same market capitalization decile at the beginning of the period. Panel A ('stocks bought') depicts stocks in the highest quintile of $\Delta\#Inst$ during the institutional trading window, separately for short-leg and long-leg anomaly stocks. Panel B ('stocks sold') similarly depicts stocks in the lowest $\Delta\#Inst$ quintile. $\Delta\#Inst$ is detrended prior to averaging across anomalies by subtracting the concurrent average cumulative $\Delta\#Inst$ of stocks in the Neutral anomaly leg (middle 40%). For momentum, the three-month anomaly return window straddles quarters 0 and 1.

Figure 2: Cumulative changes in number of institutional investors across seven anomalies

Table 8

Fama-MacBeth regressions of stock returns on institutional ownership changes. The table reports average coefficient estimates from monthly cross-sectional regressions, for the sample period July 1982 through June 2012. The dependent variable in columns 1-4 is the monthly stock return for months July t through June $t+1$. ' $\Delta\#Inst$ ' is the scaled change in number of institutions over the indicated window (in parentheses). Quarter q corresponds to the calendar quarter ending in June t . $\Delta\#Inst$ (prior quarter) is measured during the most recent calendar quarter prior to the stock return month (dependent variable). The column 4 regression includes controls for firm size, liquidity, and seven anomaly characteristics (see Table 1), observed as of fiscal year-end of calendar $t-1$ (coefficients unreported). The dependent variable in column 5 ('predicted anomaly returns') is the sum of the product of the nine control characteristics used in the column 4 regression and their corresponding coefficient estimates. The dependent variable in column 6 ('residual returns') equals total returns minus the predicted anomaly returns. All independent variables are winsorized at the 1% level in both tails. t -statistics in parentheses using Newey-West correction for serial correlation (six-month lag).

Return component:	Total			Predicted anomaly returns	Residual returns
$\Delta\#Inst (q-5;q)$	-0.50 (-3.3)				
$\Delta\#Inst (q-5;q-1)$	-0.61 (-3.7)	-0.60 (-3.7)	-0.23 (-2.5)	-0.36 (-3.0)	-0.23 (-2.5)
$\Delta\#Inst (q)$	0.29 (1.1)				
$\Delta\#Inst$ (prior quarter)		0.92 (2.0)	0.97 (2.6)	-0.08 (-0.6)	0.97 (2.6)
Characteristics	No	No	No	Yes	N/A

Table 8: Fama-MacBeth regressions of stock returns on institutional ownership changes

How to read Figure 2. The institutional buying and selling patterns persist through the anomaly return window. Stocks bought by institutions remain heavily owned by institutions; stocks sold remain relatively sold. The negative future returns therefore cannot be dismissed as returns earned after institutions already exited.

How to read Table 8. The long-horizon change in institutional ownership negatively predicts future returns. After controlling for anomaly characteristics, the residual relation remains negative. The short-horizon relation, however, remains positive. This table is the formal regression version of the horizon reconciliation.

9.6 Table 9: Fama-MacBeth regressions using alternative proxies; Table 10: Robustness checks

Table 9

Fama-MacBeth regressions using alternative proxies.

The table reports average coefficient estimates from monthly cross-sectional regressions for the sample period July 1982 through June 2012. The dependent variable is the monthly stock return for months July t through June $t+1$ (i.e., 12 regressions for each year- t institutional trading window). $\Delta\#Inst$ is the scaled change in number of institutions; $\Delta\%Inst$ ($\Delta\%MF$) is the change in percent of shares held by institutions (mutual funds); $\Delta\%NMF$ is $\Delta\%Inst - \Delta\%MF$; $\Delta\%MF(Flow)$ is the change in percent of shares held by mutual funds under in- or out-flow pressure as in [Coval and Stafford \(2007\)](#); and $\Delta\%MF(Non-flow)$ is $\Delta\%MF - \Delta\%MF(Flow)$. All change variables are measured over the six calendar quarters ending at the end of June t , and winsorized at the 1% level in both tails. Columns 7 and 8 exclude stocks in the top (bottom) decile of $\Delta\%MF(Flow)$ and in the middle three deciles of $\Delta\%MF(Non-flow)$. Column 8 includes untabulated explanatory variables including the seven anomaly characteristics, the log market capitalization, and Amihud illiquidity ratio. t -statistics in parentheses using Newey-West correction for serial correlation (six-month lag).

Sample: Column:	All						No flow pressure	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta\#Inst$	-0.50 (-3.3)		-0.54 (-3.2)			-0.66 (-2.8)	-0.67 (-3.1)	-0.29 (-2.3)
$\Delta\%Inst$		-0.61 (-2.0)	0.22 (0.8)					
$\Delta\%MF$				-0.98 (-1.3)				
$\Delta\%NMF$				-0.56 (-1.9)	-0.78 (-2.2)	0.03 (0.1)		
$\Delta\%MF$ (Flow-induced)					-0.09 (-1.8)	-0.07 (-1.5)		
$\Delta\%MF$ (Non-flow)					-0.01 (-0.7)	0.02 (0.2)		
Characteristics	No	No	No	No	No	No	No	Yes

Table 9: Fama-MacBeth regressions using alternative proxies

Table 10

Robustness checks.

This table reports the average coefficient estimates from Fama-MacBeth monthly stock return regressions as in [Table 7](#), on subsamples. In Panel A the subsamples exclude stocks with (i) fewer than 20 institutional investors as of the end of June of t , (ii) a market capitalization below the 20th percentile of NYSE stocks as of June of year t , or (iii) a market capitalization below NYSE median as of June of year t . In Panel B the subsamples are by portfolio formation year: 1982–1996 and 1997–2011. t -statistics in parentheses are estimated using Newey-West serial correlation-consistent standard errors with a six-month lag.

Panel A: Fama-MacBeth regressions excluding small stocks							
Component:	Predicted anomaly returns			Residual returns			
	Sample:	#Inst ≥ 20	>NYSE 20%	>NYSE 50%	#Inst≥20	>NYSE 20%	>NYSE 50%
Δ#Inst (q-5;q-1)		-0.40 (-3.2)	-0.38 (-2.5)	-0.47 (-2.5)	-0.24 (-2.4)	0.01 (0.1)	-0.10 (-0.5)
Δ#Inst (prior quarter)		0.02 (0.9)	0.09 (0.6)	0.18 (0.9)	1.18 (2.7)	0.72 (1.2)	0.09 (0.1)
Panel B: Fama-MacBeth regressions in subperiods							
Component:	Predicted anomaly returns			Residual returns			
	Subperiod:	1982-1996	1997-2011	1982-1996	1997-2011		
Δ#Inst (q-5;q-1)		-0.28 (-2.9)	-0.43 (-2.0)	0.04 (0.4)	-0.49 (-3.3)		
Δ#Inst (prior quarter)		0.06 (0.6)	-0.21 (-0.9)	1.15 (4.2)	0.80 (1.1)		

Table 10: Robustness checks

How to read Table 9. The negative long-horizon relation is not driven by mutual fund flows. The $\Delta\#Inst$ measure remains negative when the authors control for changes in institutional percent ownership, mutual fund ownership, flow-induced mutual fund trading, and when they exclude stocks under flow pressure.

How to read Table 10. The results are not only a micro-cap phenomenon and are not confined to the early anomaly-discovery period. The anomaly-predicted component remains negatively related to institutional demand even among larger stocks and in both subperiods.

10 Detailed table-by-table takeaway

Visual	Main takeaway
Table 1	The paper chooses seven representative anomalies across broad categories, creating a clean test of whether institutions respond to well-known predictors of returns.
Table 2	The anomaly returns are economically meaningful during the sample, and the average abnormal return is concentrated in the short leg.
Table 3	Institutional ownership is large and variable enough to study changes in institutional investor base as an active demand signal.
Figure 1	The empirical design measures institutional trading before anomaly returns, not after the anomaly has already paid off.
Table 4	Institutions often buy short-leg stocks more than long-leg stocks, rejecting the most basic prediction of the sophisticated-institution hypothesis.

Visual	Main takeaway
Table 5	Short-leg stocks bought by institutions have the most negative future alphas; institutional buying does not identify short-leg stocks that avoid the anomaly.
Table 6	The conditional portfolios are not obviously explained away by simple differences in institutional breadth, market cap, volatility, or illiquidity.
Table 7	The sign of the institutional demand-return relation depends on horizon: short-term positive, long-term negative.
Figure 2	Institutional positions persist, so the long-horizon underperformance remains relevant for institutional portfolios.
Table 8	Fama-MacBeth regressions show both adverse anomaly exposure and a negative residual institutional-demand effect.
Table 9	Mutual fund flow does not explain the main results.
Table 10	The institutional preference for anomaly-short-leg stocks is robust across larger-stock samples and subperiods.

11 Critical assessment

11.1 Strengths

- The paper asks a simple, sharp question: do institutions exploit anomalies?
- The timing design is careful and directly tied to anomaly portfolio formation.
- The results are robust across institutional-demand measures, flow controls, size filters, and subperiods.
- The horizon analysis provides an important reconciliation with the prior literature that finds positive short-run effects of institutional demand.

11.2 Limitations

- The analysis is aggregate. Some institutional categories may exploit anomalies even if aggregate institutional demand does not.
- The paper does not fully identify the exact motive behind institutional buying of overvalued stocks.
- The evidence is stronger for rejecting simple arbitrageur interpretations than for proving one specific behavioral or agency mechanism.
- The paper studies gross portfolio returns; real-world performance also depends on implementation costs, mandates, and shorting constraints.

11.3 Best use of the paper

Use this paper when discussing:

- institutional investors and market efficiency;
- limits of arbitrage;
- stock return anomalies;
- institutional herding and demand pressure;
- agency problems in delegated asset management;
- the distinction between short-horizon and long-horizon return predictability.

12 One-paragraph summary

Edelen, Ince, and Kadlec show that institutional investors do not generally trade in the direction prescribed by stock return anomalies. Instead, aggregate institutional demand often increases in short-leg stocks classified as overvalued, and these stocks subsequently earn especially negative abnormal returns. The negative relation between institutional demand and future returns is a long-horizon phenomenon that coexists with the positive short-horizon relation documented in prior work. The results are not explained by mutual fund flows and challenge standard views of institutions as arbitrageurs constrained only by frictions. The paper suggests that institutions may sometimes contribute to anomaly mispricing rather than eliminate it.